The KIR BMRC shared research facility for high performance computing

Konstantin Borisenko, Jonathan Truslow and Stephen Sansom, 31st July 2023

Table of Contents

1. Introduction ***1***

2. Documentation ***2***

3. Arranging access ***2***

4. Current access charges ***3***

4.1 Future costings (for grant applications) **3**

4.2 Group cost deficit charge **3**

4.3 Explanation of access charges **3**

4.3.1 Status 3

4.3.2 Resource needs: 4

4.3.3 Frequency of use: 4

4.3.4 Working hours and contract duration 4

1. Introduction

The KIR BMRC SRF provides a HPC platform for KIR users. It is hosted on the Biomedical Research Computing (BMRC) platform, which it extends to provide additional capabilities and support for our researchers. It is designed and operated to enable the KIR to meet Information Governance requirements for data science research.

The University's BMRC platform provides access to ~7000 compute cores, 60 GPU cards,10PB of high-performance storage and advanced cloud capabilities. The system runs CentOS-7 and uses the SLURM job scheduler. It is supported by a core team of system administrators led by Dr Robert Esnouf.

Building on the foundational BMRC capabilities, the KIR BMRC SRF additionally provides:

- A dedicated systems administrator, Dr Konstantin Borisenko.
- A Kennedy stack of compatible software modules.
- Dedicated spaces for shared files (including for community datasets and collaborative projects) and to support the activities of other KIR SRFs.
- Persistent access to Rstudio and Jupyter notebooks via a dedicated Posit workbench server.
- Access to a large number of BMRC compute shares.

The facility is part of the KIRs Data Science Platform and is overseen by Associate Professor Stephen Sansom.

2. Documentation

Documentation for the underlying BMRC can be found here: the Biomedical Research Computing (BMRC) platform.

Documentation for users of the KIR BMRC SRF is available here: https://rcwiki.kennedy.ox.ac.uk.

3. Arranging access

In the first instance please contact the KIR systems administrator konstantin.borisenko@kennedy.ox.ac.uk.

Following completion of an access request form, an induction to the system will be arranged.

Please note that **provision of a grant code is required**, and this will be charged before access to the system is granted.

4. Current access charges

Annual costs shown; these apply from 1st August 2023 – 31st July 2024.

		Resource needs		
		standard	**compute only**	**access only**
Student	occasional use	£1,000	£500	£0
	routine use	£2,000	£1,000	£0
PDRA	occasional use	£3,000	£1,500	£300
	routine use	£6,000	£3,000	£300
PI	occasional use	£1,200	£600	£300
	routine use	£6,000	£3,000	£300
Group	cost deficit charge	*Determined on a case-by-case basis if applicable*		

4.1 Future costings (for grant applications)

Costs are reviewed annually. They are expected to increase to track internal BMRC pricing and inflation. **Please allow for minimum yearly increases of 5%**.

4.2 Group cost deficit charge

If a research group incurs storage and/or compute costs significantly greater than the total access charge(s) recovered from the group, the deficit may be charged to the group at the end of the year.

This is, for example, likely to apply when a group has no active users but is keeping data on the system.

4.3 Explanation of access charges

KIR BMRC SRF access charges are partially subsidised by KTRR core funding provided to the Data Science Platform to support and promote computational research within the KIR.

The SRF charges users according to their:

a) **status**: “student”, “PDRA”, “PI”, “visitor”
b) **resource needs**: “standard”, “compute only” or “access-only”
c) **frequency of usage:** “routine” or “occasional”
d) **working hours and contract duration**

Each of these is explained below:

4.3.1 Status

- **Students**: Access for students is most heavily subsidised (i) to promote training and (ii) due to the limited funds associated with studentships.
- **Post-docs:** Access charges for PDRAs more closely reflect actual costs and must be written into grant costings.

- **PI:** PIs typically need access for occasional group management/supervision activities and less frequently for developing/running data analysis workflows. Occasional charges are set accordingly. However, for PIs who are using the cluster to perform work routinely, costs are the same as for PDRAs.
- **Visitors/Oxford staff outside KIR:** access is provided at the discretion of the SRF management team based on compatibility with KIR Information Governance requirements. A normal KIR visitor/access agreement with an MSD IT log-in account must be in place. Costs for access are the same as for the equivalent KIR users.

4.3.2 Resource needs:

- **Standard**: This is appropriate for e.g. genetic, genomic, image analysis and machine learning/data science research etc.

- **Access only**: for researchers who only need access to data stored on the system or who wish to explore/learn how to use the system. Individual users' disk and CPU usage are reviewed quarterly. If such users are likely to use more >100GB disk space and/or start to perform data processing, we will discuss future charges with the relevant user + PI.

- **Compute only***:* on a case-by-case basis this may be agreed for specific types of HPC work* which require access to CPUs but not significant storage (<100GB of persistent storage across a quarter; temporary KIR scratch space can be used without penalty). *Specific types of work include:

    - Mathematical modelling projects.
    - Image analysis projects where data is stored on a linked platform for which costs have been charged separately.

4.3.3 Frequency of use:

- **Routine***:* for researchers who will regularly use the KIR BMRC SRF.

- **Occasional**: for researchers conducting both experimental and computational research, where the computational aspect of the individual's research activities comprises ≤ 50% of their efforts.

- **"One-off" access:** For full-time students and staff, one-off "routine" access charges for a period of up to 3 months may be prorated at a fixed 25% rate. "Access only" and "occasional" charges will not be prorated for these users. "One-off" access may be requested once every 3 years. If further access is required within a three-year period, normal yearly costs will be applied (less the initial charge if within the same financial year).

4.3.4 Working hours and contract duration

Full time staff and students will be charged as described above.

For (a) part-time users, (b) visitors/summer students on short term contracts/appointments or (c) employees whose contracts are coming to their end in the current financial year "occasional" and "routine" access charges may be prorated at 25%, 50% or 75% of the normal yearly cost.

Please note that "access-only" fees cannot be prorated for any users.